

# DAY 9: Set operations and methods

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→ Set is unordered collection of elements no duplicate allowed in the set and set is represented by { }.

Eg:

```
nums = { 1,2,3,4}
```

```
print(nums)
```

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## Operations

union ( )

→ The union ( ) will combine two sets into a single set

syntax → set\_1.union(set\_2) or set\_1 | set\_2

Eg:

```
nums={1,2,3,2}
```

```
data={5,6}
```

```
print(nums.union(data))
```

```
nums={1,2,3,2}
```

```
data={5,6}
```

```
print(nums|data)
```

Intersection()

→ This will gives us the common elements from both sets

syntax → set\_1.intersection(set\_2) or set\_1 & set\_2

Eg:

```
nums={1,2,3,4}
```

```
data={4,5,6}
```

```
print(nums.intersection(data))
```

```
print(nums&data)
```

difference ( )

→ It will display the different elements from set\_1 but not set\_2 elements

syntax → `set_1.difference(set_2)` or `set_1-set_2`

Eg:

```
nums={1,2,3,4}
data={4,5,6}
print(nums.difference(data))
print(nums-(data))
```

`symmetric_difference ( )`

→ different elements from the both sets

syntax → `set_1.symmetric_difference(set_2)` or `set_1 ^ set_2`

Eg:

```
nums={1,2,3,4}
data={4,5,6}
print(nums.symmetric_difference(data))
print(nums^data)
```

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## Methods

`add()`

→ `add ( )` method will add only one element at a time

syntax → `set.add(element)`

Eg:

```
data={1,2,3,4}
print(data)
data.add(7)
print(data)
```

`update()`

→ `update ( )` method will Add multiple elements at a time

syntax → `set.update([elements])` or `set_1.update(set_2)`

Eg:

```
data={1,2,3,4}
print(data)
```

```
data.update([7,9])
print(data)

nums={1,2,3,4}
data={4,5,6}
data.update([7,8])
print(data)
data.update(nums)
print(data)

remove()
```

→ remove () method will delete the given element from the set, if the element is not present in the set, it will raise error

syntax→ set.remove(element)

Eg

```
data={1,2,3,4}
print(data)
data.remove(2)
print(data)

discard()
```

→ The method is used to del the elements from the set, but never raise any error even the element not inside set

syntax→ set.discard(element)

Eg:

```
data={1,2,3,4}
print(data)
data.discard(1)
print(data)
data.discard(5)
print(data)

clear()
```

→ The method is used to del all elements from the set and it will written empty set

syntax→ set.clear()

Eg:

```
data={1,2,3,4}  
print(data)  
data.clear()  
print(data)
```